MACHINE LEARNING • LINEAR MODELS

RIDGE REGRESSION

How a linear model balances prediction accuracy and complexity by penalizing large coefficients.

What happens when Ordinary Least Squares coefficients grow too large?

OLS estimates coefficients without constraints.

LIMITATIONS OF OLS

Minimizes SSE Only
Ignores coefficient magnitude entirely.

Noise Sensitivity
Large coefficients cause the model to overfit training data.

Multicollinearity
Highly correlated features create unstable coefficients.

Ridge adds a structural penalty.

THE RIDGE SOLUTION

Applies a Penalty
Discourages excessively large weights.

Balances Tradeoff
Optimizes prediction error vs. coefficient size.

Reduces Overfitting
Improves generalizability on unseen data.

02 • RIDGE OBJECTIVE

How does Ridge Regression change the optimization goal?

It adds an L2 penalty to the original OLS loss function.

$$\text{Loss} = \text{SSE} + \alpha \times \Sigma w_j^2$$

Add squared coefficients to the squared prediction errors.

WHY PENALIZE THE WEIGHTS?

- Prevents any single feature from dominating.
- Reduces model sensitivity to noise.

SSE (SUM OF SQUARED ERRORS)

Measures the difference between actual and predicted values.

A (ALPHA)

Regularization strength. Controls how strongly large coefficients are penalized.

ΣW_j^2 (L2 PENALTY)

Cost based on squared feature coefficients. Standard intercept is not penalized.

03 • NUMERICAL EXAMPLE

Calculating the Ridge loss for a specific model configuration.

Evaluate the penalty and total objective.

INITIAL MODEL

$$\hat{y} = 10 + 100x_1 + 80x_2$$

GIVEN PARAMETERS

OLS SSE = 500

Alpha (α) = 0.1

$w_1 = 100$

$w_2 = 80$

CALCULATE L2 PENALTY

Penalty = 0.1 × (100² + 80²)

Penalty = 0.1 × (10,000 + 6,400)

Penalty = 1,640

TOTAL RIDGE LOSS

Ridge Loss = SSE + Penalty

Ridge Loss = 500 + 1,640

Ridge Loss = 2,140

** The penalty is a mathematical cost to constrain weights, not a literal prediction error.*

04 • OPTIMIZATION

How can the weights be updated using the Ridge gradient?

Gradient Descent moves parameters toward a lower total Ridge objective.

> **New weight = Old weight − η × Gradient**
>
> η (eta) is the learning rate.

ILLUSTRATIVE UPDATE (A = 0.1)

PARAMETER	OLD VALUE	GRADIENT	UPDATED VALUE
w_0	10	60	**9.40**
w_1	100	46	**99.54**
w_2	80	46	**79.54**

EXAMPLE: H = 0.01

$w_0 = 10 - 0.01(60) = 9.40$

$w_1 = 100 - 0.01(46) = 99.54$

$w_2 = 80 - 0.01(46) = 79.54$

UPDATED MODEL

$$\hat{y} = 9.40 + 99.54x_1 + 79.54x_2$$

The gradient contains both the data-error and L2 penalty.

05 • RECAP

Ridge Regression: the key ideas to remember

MODEL

$$\hat{y} = w_0 + w_1 x_1 + \dots$$

Weighted combination of input features.

PENALTY

$$L2 = \alpha \times \Sigma w_j^2$$

Cost added to constrain large coefficients.

GOAL

minimize SSE + L2

Balance prediction error and model complexity.

SCIKIT-LEARN IMPLEMENTATION

```
from sklearn.linear_model import Ridge

model = Ridge(alpha=0.1)
model.fit(X, y)
```

REMEMBER

Ridge reduces overfitting but does not guarantee better test performance. Feature scaling is essential.

NEXT →

Lasso Regression • Elastic Net